Civic.ai: Bangla-English RAG for Bangladeshi Government Services

Final Year Thesis - Phase 2 Design, Alternatives, and Functional Verification

Current Domain: Birth and Death Registration

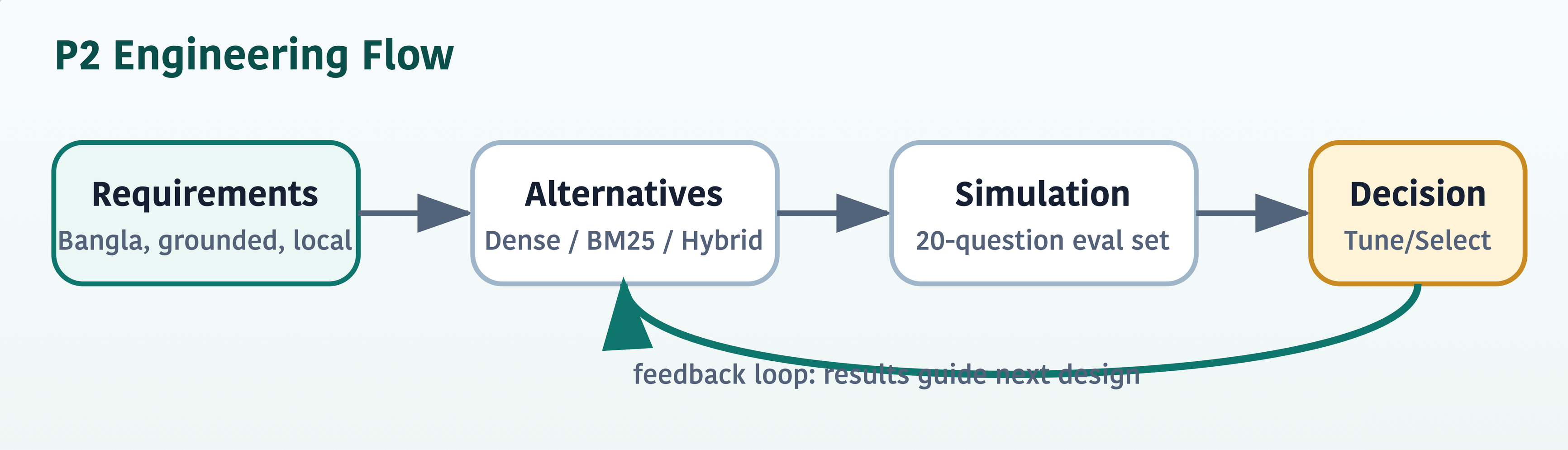
Problem and Motivation

Government-service information in Bangladesh is scattered across PDFs, HTML pages, notices, FAQs, tables, legal rules, and scanned or converted documents.

Goal: build a factual and explainable RAG chatbot that answers Bangla-English government-service questions using retrieved evidence.

- Accuracy and grounding are more important than fluency.
- Bangla and code-mixed queries must be supported.
- Answers should include source IDs and official/source links.

Phase 2 Objective Mapping



- **Design process:** modular RAG pipeline from data cleaning to answer generation.
- **Alternatives:** Dense-only, BM25-only, Hybrid RRF, Hybrid+Rerank, Simple RAG, CivicRAG.
- **Verification:** 20-question Bangla evaluation set with expected source chunks.
- **Constraints:** local LLMs, no API-key dependency, noisy Bangla government data.

Data Preparation

Data Type	Processing Choice
FAQ JSON	One question-answer pair per chunk
Legal rules/acts	Section or rule-based chunks
Application process	Heading and step-based chunks
Fee tables	Table-level and row-level chunks
Noisy HTML/PDF	Manual/semi-manual cleaning and metadata

Key design: separate `retrieval_text` for search from original source content for answer grounding.

CivicRAG Pipeline

Raw Government Data

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Cleaning + Normalization

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Document-Type-Aware Chunking + Metadata

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BGE-M3 Embeddings + ChromaDB

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User Query: Bangla / English / Code-mixed

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Dense Search + BM25 Search

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RRF Fusion + Reranking / Intent Boosting

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Safe Answer Path or Local LLM Fallback

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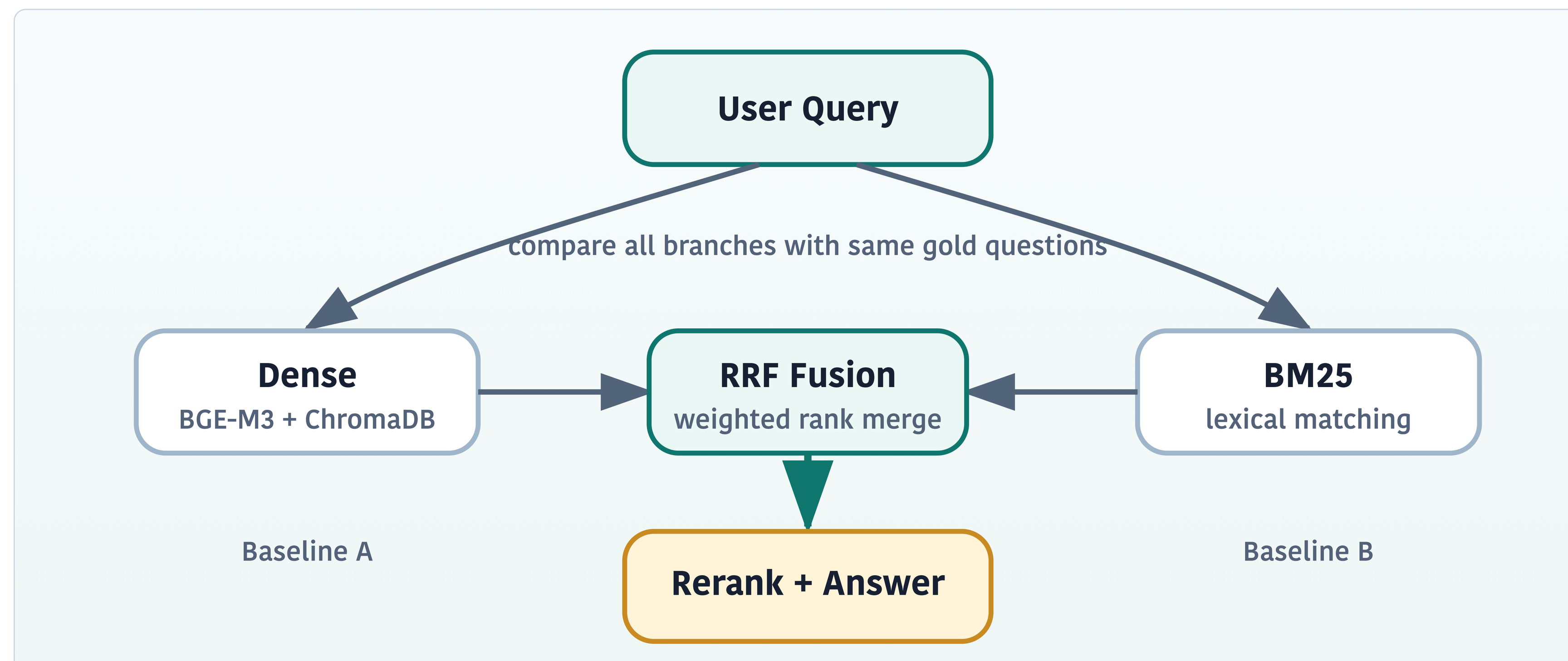
Final Answer + Source IDs + Official Links

Retrieval Alternatives



1. **Dense-only:** BGE-M3 embedding search in ChromaDB.
2. **BM25-only:** sparse lexical matching over Bangla tokens.
3. **Hybrid RRF:** rank fusion of Dense and BM25 results.
4. **Hybrid + Rerank:** rerank the fused candidate set.

$$\mathrm{RRF}(d) = \Sigma_{r \in R} w_r / (k + \mathrm{rank}_r(d))$$

RRF rewards chunks that appear high in one or both retrievers; it does not only keep common chunks.

Evaluation Metrics

$$\mathrm{Recall@}k = 1[R_k \cap G \neq \varnothing]$$

$$\mathrm{Context\ Precision@}k = |R_k \cap G| / k$$

$$\mathrm{MRR} = \mathrm{average}(1 / \text{rank of first relevant chunk})$$

$$\mathrm{nDCG@}k = \mathrm{DCG@}k / \mathrm{IDCG@}k$$

Current P2 Retrieval Results

Evaluation set: 20 Bangla scenario/paraphrase questions with expected source chunk IDs.

Method	R@1	R@3	R@5	MRR
BM25-only	0.450	0.550	0.800	0.546
Dense-only	**0.800**	**0.900**	**0.900**	**0.842**
Hybrid RRF	0.700	0.800	0.850	0.752
Hybrid+Rerank	0.650	0.800	0.850	0.738

Finding: Dense-only currently performs best. Hybrid retrieval remains a candidate, but requires tuning before being selected as final.

Paraphrase Robustness Finding

Same-intent questions should retrieve and answer consistently.

> জন্ম নিবন্ধন হারালে কী করব?

> আমার জন্মসনদ হারিয়ে গেছে, নতুন কপি কোথায় পাব?

Testing showed that retrieval can be correct while answer routing can still fail. This motivates separate evaluation for retrieval correctness and answer correctness.

Next Steps

- Expand gold evaluation set to 50-100 reviewed questions.
- Add paraphrase clusters for each civic intent.
- Tune RRF weights and top-k values.
- Run full cross-encoder reranking on selected subsets.
- Add manual scoring for faithfulness, correctness, completeness, and language quality.
- Add domain routing before Passport/BRTA expansion.

Artifacts

CivicRAG BGE-M3 ChromaDB BM25 RRF Local LLMs

Repository: github.com/Shihabsarker93/civic_ai_rag